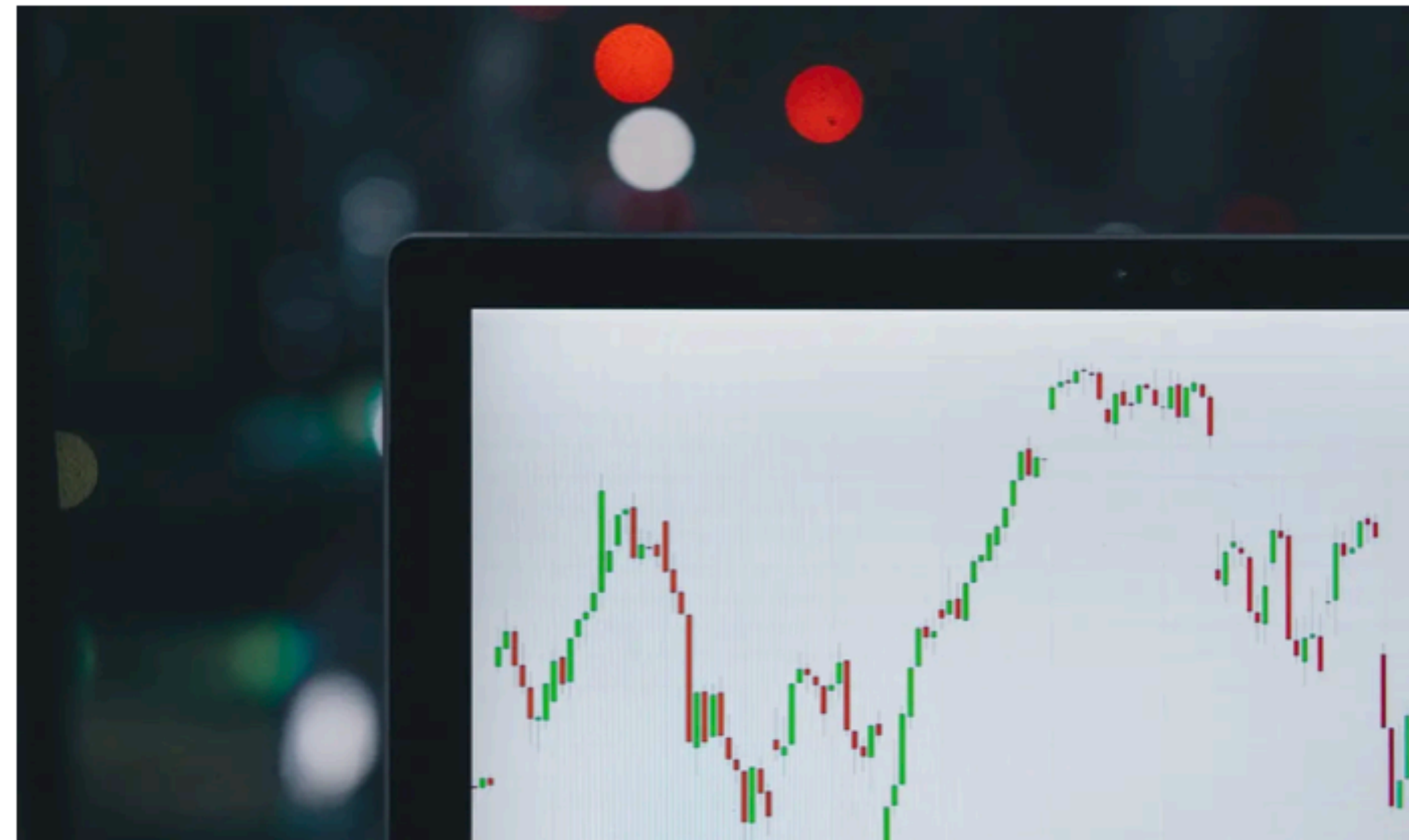


Big Data lab - OLTP

Stock market data

Finnhub

Real-time Market Data



Finnhub

BTC-USD	70771.93	ETH-USD	2188.5	EUR/USD	1.1688	GBP/USD	1.3428	AAPL	258.83	MSFT	369.32	AMZN	
^ 0.02% (16.58)		▼ -0.15% (-3.30)		▼ -0.32% (-0)		▼ -0.24% (-0)		▼ -0.64% (-1.66)		▼ -1.01% (-3.75)		^ 0.6	

Global Filings, Transcripts, and Presentations data are available on Finnhub API. Check out [here](#)

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BTC-USD	70778.74	ETH-USD	2188.79	EUR/USD	1.1689	GBP/USD	1.343	AAPL	258.9	MSFT	369.48	AMZN
↑ 0.03% (23.39)		↓ -0.14% (-3.01)		↓ -0.31% (-0)		↓ -0.22% (-0)		↓ -0.61% (-1.59)		↓ -0.96% (-3.59)		↑ 0.7

Global Filings, Transcripts, and Presentations data are available on Finnhub API. Check out [here](#)

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API Key

Email: superk@cs.huji.ac.il

.....rt7490



API Documentation

Regenerate

Finnhub



Open Data

WEBSOCKET

Trades

High Usage

News

Premium

Press Releases

Premium

STOCK FUNDAMENTALS

Symbol Lookup

Stock Symbols

Market Status

New

Market Holiday

New

Company Profile

Premium

Company Profile 2

New

Company Executive

Premium

Market News

Company News

High Usage

Press Releases

Premium

News Sentiment

Premium

Peers

Basic Financials

High Usage

Ownership

Premium

Fund Ownership

Premium

Trades - Last Price Updates

Stream real-time trades for US stocks, forex and crypto. Trades might not be available for some forex and crypto exchanges. In that case, a price update will be sent with volume = 0. A message can contain multiple trades. 1 API key can only open 1 connection at a time.

The following FX brokers do not support streaming: FXCM, Forex.com, FHFx. To get latest price for FX, please use the Forex Candles or All Rates endpoint.

Method: `Websocket`

Examples:

```
wss://ws.finnhub.io
```

Response Attributes:

type

Message type.

data

List of trades or price updates.

s

Symbol.

p

Last price.

t

UNIX milliseconds timestamp.

v

Volume.

Sample code

```
1 #https://pypi.org/project/websocket_client/
2 import websocket
3
4 def on_message(ws, message):
5     print(message)
6
7 def on_error(ws, error):
8     print(error)
9
10 def on_close(ws):
11     print("### closed ###")
12
13 def on_open(ws):
14     ws.send('{"type":"subscribe","symbol":"AAPL"}')
15     ws.send('{"type":"subscribe","symbol":"AMZN"}')
16     ws.send('{"type":"subscribe","symbol":"BINANCE:BTCUSDT"}')
17     ws.send('{"type":"subscribe","symbol":"IC MARKETS:1"}')
18
19 if __name__ == "__main__":
20     websocket.enableTrace(True)
21     ws = websocket.WebSocketApp("wss://ws.finnhub.io?token=d0596
22                               on_message = on_message,
23                               on_error = on_error,
24                               on_close = on_close)
25     ws.on_open = on_open
26     ws.run_forever()
27
```

Sample response

```
1 {
2   "data": [
3     {
4       "p": 7296.89,
5       "s": "BINANCE:BTCUSDT",
6       "t": 1575526691134,
7       "v": 0.011467
8     }
9   ]
10 }
```

Variable to calculate

<code>symbol</code>	<code>(string: "AMZN", "AAPL", ...)</code>
<code>data_time</code>	<code>(time YYYY-mm-ddTHH:MM:SS.msZ)</code>
<code>now</code>	<code>(time YYYY-mm-ddTHH:MM:SSZ)</code>
<code>EMA</code>	<code>(number: \$USD)</code>
<code>min</code>	<code>(number: \$USD)</code>
<code>max</code>	<code>(number: \$USD)</code>
<code>count</code>	<code>(integer)</code>
<code>mean</code>	<code>(average)</code>
<code>ss</code>	<code>(number: sum of squares)</code>
<code>var</code>	<code>(number: variance)</code>
<code>smp1_var</code>	<code>(number: variance)</code>

Variable to calculate

```
symbol      = META
data_time   = 2026-04-13T19:01:35.959Z
now         = 2026-04-13T19:02:14Z
EMA         = 626.907215800003
min         = 626.88
max         = 627.62
count       = 39
mean        = 627.355256410256
ss          = 0.953247435897588
var         = 0.024442241946092
smp1_var    = 0.0250854588394102
```

$$SMA_k = \frac{p_{n-k+1} + p_{n-k+2} + \cdots + p_n}{k}$$

$$= \frac{1}{k} \sum_{i=n-k+1}^n p_i$$

$$SMA_{k,\text{next}} = \frac{1}{k} \sum_{i=n-k+2}^{n+1} p_i$$

$$= \frac{1}{k} \left(\underbrace{p_{n-k+2} + p_{n-k+3} + \cdots + p_n + p_{n+1}}_{\sum_{i=n-k+2}^{n+1} p_i} + \underbrace{p_{n-k+1} - p_{n-k+1}}_{=0} \right)$$

$$= \frac{1}{k} \left(\underbrace{p_{n-k+1} + p_{n-k+2} + \cdots + p_n}_{=SMA_{k,\text{prev}}} \right) - \frac{p_{n-k+1}}{k} + \frac{p_{n+1}}{k}$$

$$= SMA_{k,\text{prev}} + \frac{1}{k} \left(p_{n+1} - p_{n-k+1} \right)$$

Formula for Exponential Moving Average (EMA)

$$EMA_{\text{Today}} = \left(\text{Value}_{\text{Today}} * \left(\frac{\text{Smoothing}}{1 + \text{Days}} \right) \right) + EMA_{\text{Yesterday}} * \left(1 - \left(\frac{\text{Smoothing}}{1 + \text{Days}} \right) \right)$$

